

PRODUCT FEASIBILITY REPORT

PILOT SCALE TALBINA MANUFACTURING PLANT

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Delivered to: Sir Anjum

PART A: PILOT SCALE FEASIBILITY

(20 kg per day – 8 hour single shift)

1. Plant Capacity & Production Assumptions

Parameter	Details
Product	Talbina (powder form)
Daily Capacity	20 kg/day
Shift	1 shift (8 hours)
Monthly Working Days	25
Monthly Production	500 kg
Packaging Size	250 g pouches
Monthly Packs	2,000 packs

2. Product Formulation (20kg batch)

Ingredient	Quantity (kg)
Barley flour	13.0
Dates	4.0
Chia seeds	1.0
Sesame seeds	1.0
Olive oil	0.6
Cinnamon	0.2
Vanilla extract	0.2
Total	2,000 packs

3. Manufacturing Process

1. Raw material inspection
2. Sorting and washing of fresh dates
3. Preparation of date paste
4. Grinding of dates & chia (if whole)
5. Roasting/drying of barley flour
6. Mixing all ingredients
7. Sieving for uniformity
8. Weighing & packaging
9. Storage & dispatch

4. Machinery & Equipment Requirement (Pilot Scale)

Equipment	Capacity	Power	Estimated Cost (PKR)
Hammer mill / grinder	20–30 kg/hr	3KW	180,000
Ribbon mixer	25 kg/batch	2KW	220,000
Roasting / drying unit	Small batch	4KW	150,000
Vibro sifter	Small	1KW	80,000
Digital weighing scale	-	-	25,000
Sealing machine	Semi-automatic	1.5KW	120,000
Stainless steel tables & tools	-	-	60,000
Total		11.5KW	835,000 PKR

5. Utilities Requirement

Utility	Estimated Cost (PKR/month)
Electricity	120,000
Water	3,000
Gas (if roasting)	7000
Total Utilities	130,000 pkr

Electricity Consumption: Assume average running load = 9KW
(Industrial plants rarely run 100% load continuously)

- Per day units = $9 \times 8 = \mathbf{72 \text{ units}}$
- Monthly units = $72 \times 25 = \mathbf{1800 \text{ units}}$
- Industrial rate = 68 PKR/unit (4x) = **120,000 Pkr/month**

6. Packaging details

Item	Monthly Cost (PKR)
Printed pouch (250 g	36,000
Label / coding	8,000
Cartons & handling	6,000
Total Packaging Cost	50,000 PKR

6. Plant Rent & Area

Item	Details
Covered Area	500–600 sq.ft
Location	Semi-industrial area
Monthly Rent	25,000 PKR

7. Labor Cost (Monthly)

Staff	Cost
Production worker	22,000
Helper 1	18,000
Supervisor (part-time)	15,000
Total Labor Cost	55,000

7. Total Project Cost (Pilot Scale - Monthly)

Cost Head	PKR
Raw materials	275,000
Packaging	50,000
Utilities	130,000
Labor	55,000
Rent	25,000
Miscellaneous	12,000
Total Monthly Cost	547,000

“Machinery cost is one time capital investment; therefore, it has been included in total project cost and not in monthly operating cost.”

8. Cost of Production (Pilot Scale)

Cost per kg = 890pkr

Cost per 250g pack = 223pkr

9. Profit Margin (Pilot Scale)

- 30% Profit margin
- Selling price with 30% profit = **290pkr per 250 g pack**

PART B: LARGE SCALE FEASIBILITY

1. Plant Capacity & Production Assumptions

Parameter	Details
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Product	Talbina (powder form)
Daily Capacity	200 kg/day
Shift	1 shift
Monthly Working Days	25
Production rate	25kg/hour
Monthly Production	5000 kg
Packaging Size	250 g
Packs per hour	100 packs/hour
Packs per day	800 packs/day
Monthly Packs	20,000 packs

3. Product Formulation (20kg batch)

Ingredient	Quantity (kg)
Barley flour	130
Dates	40
Chia seeds	10
Sesame seeds	10
Olive oil	6
Cinnamon	2
Vanilla extract	2
Total	200 kg

3. Manufacturing Process

4. Raw material inspection
5. Sorting and washing of fresh dates
6. Preparation of date paste
4. Grinding of dates & chia (if whole)
5. Roasting/drying of barley flour
6. Mixing all ingredients
7. Sieving for uniformity
8. Weighing & packaging
9. Storage & dispatch

4. Machinery & Equipment Requirement (Pilot Scale)

Equipment	Capacity	Power	Estimated Cost (PKR)
Industrial grinder	200 kg/hr	15KW	750,000
Date paste machine	200 kg/hr	8KW	450,000
Ribbon mixer	500 kg/hr	10KW	1,200,000
Roasting / drying unit	250 kg/hr	20KW	900,000
Vibro sifter	-	5KW	450,000
Automatic packing machine	40 packs/min	8KW	1,800,000
Conveyor & handling	-	4KW	350,000
QC equipment	-	-	200,000
Total			5,300,000 PKR

5. Utilities Requirement

Utility	Estimated Cost (PKR/month)
Electricity	840,000
Water	20,000
Gas (if roasting)	80,000
Total Utilities	940,000pkr

Electricity Consumption: Assume average running load = 80KW
(Industrial plants rarely run 100% load continuously)

- Per day units = $60 \times 8 = 480$ units
- Monthly units = $480 \times 25 = 12,000$ units
- Industrial rate = 70 PKR/unit (4x) = **840,000pkr/month**
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6. Packaging details

Item	Monthly Cost (PKR)
Printed pouch	300,000
Cartons & handling	60,000
Total Packaging Cost	360,000 PKR

6. Plant Rent & Administration

Item	PKR
Factory rent (3000 sq.ft)	90,000
Admin & Maintenance	50,000

7. Total Project Cost (Pilot Scale - Monthly)

Cost Head	PKR
Machinery Cost	5,300,000
Raw materials	2,480,000
Packaging	360,000
Utilities	940,000
Labor	450,000
Rent	90,000
Miscellaneous	50,000
Total Monthly Cost	9,000,000 PKR

8. Cost of Production (Large Scale)

Cost per kg = 874pkr

Cost per 250g pack = 218pkr

9. Profit Margin (Large Scale)

- 30% Profit margin
- Selling price with 30% profit = **280-300pkr per 250 g pack**